



Wealth Over Water

A Visual Analysis of Water Security and Environmental
Governance Impacts of Illegal Gold Mining in Ghana

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Galamsey:

/gah-lam-say/

A Ghanaian term originating from “gather and sell,” historically referring to small-scale gold mining. Today, it is commonly used to describe illegal or unregulated small-scale mining operations.

These activities have been linked in multiple studies and government assessments to environmental degradation, including loss of arable land, deforestation, and contamination of rivers with heavy metals such as mercury, lead, and arsenic.

Galamsey remains a central issue in national discussions on environmental regulation, economic sustainability, and the governance of Ghana’s mineral resources.

Wealth Over Water

How Illegal Mining is Polluting Ghana's Waters

Ghana, Africa's top gold producer, is facing a significant water contamination challenge linked to illegal mining, with studies estimating that 60% of the country's fresh water bodies show unsafe levels of pollution. Elevated concentrations of toxic metals in several regions pose risks to long-term water security, and emerging research is examining potential associations with health and neonatal complications in affected communities.

2030

Ghana faces potential water import dependency by 2030 if the crisis is not resolved.

Arsenic (As)

Levels in contaminated rivers up to **330×** above national standards. Linked to **cancers** and **skin disease**.

Lead (Pb)

Concentrations in water near mining sites as high as **30×** the WHO safe limit. **Damages the brain and nervous system, especially in children.**

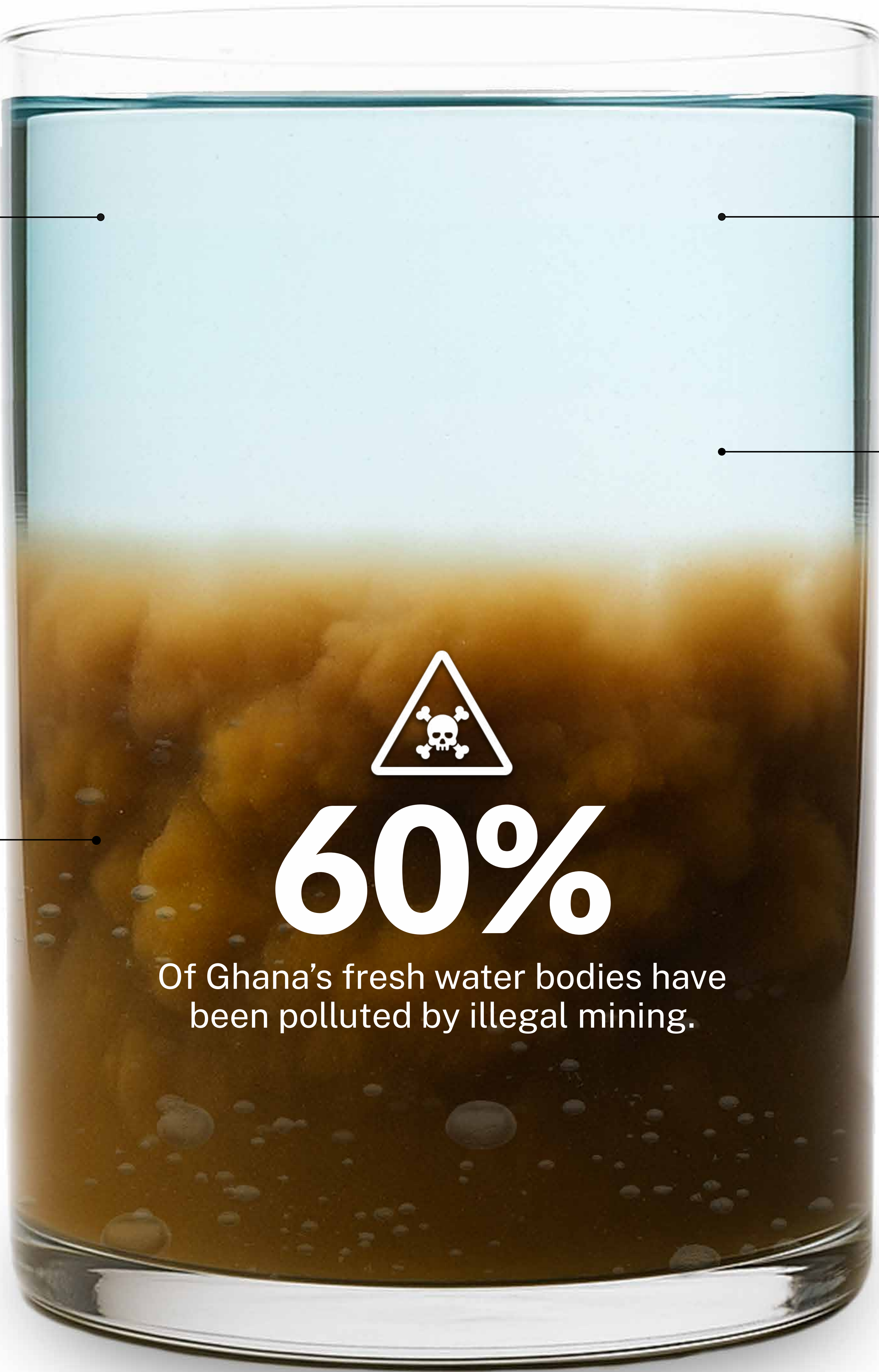
Mercury (Hg)

Found in water near mining sites at up to **30×** the WHO safe limit. Effects include **neurological, kidney, heart, and lung damage.**

SDGs Under Threat



COCOBOD, Pure Earth, WHO, Ghana EPA, Ghana Water Company, Gold Water Benchmarking 2024, Water Resources Commission, PubMed



50% Wasted

Extreme turbidity now forces Ghana's Water Company to waste up to 50% of the water it treats, up from a previous normal of only **5%**.

21,000 l / Oz

Producing 1 ounce of gold consumes around 21,000 litres of water - equivalent to a person's drinking supply for over **20 years**.

This is likely higher in galamsey due to inefficient machinery and environmentally harmful practices.

\$2+ Billion Lost

Ghana loses over \$2 billion annually to galamsey and gold smuggling.



More People, Less Water

Per-capita renewable water availability has fallen by nearly 50%` since the 1960s

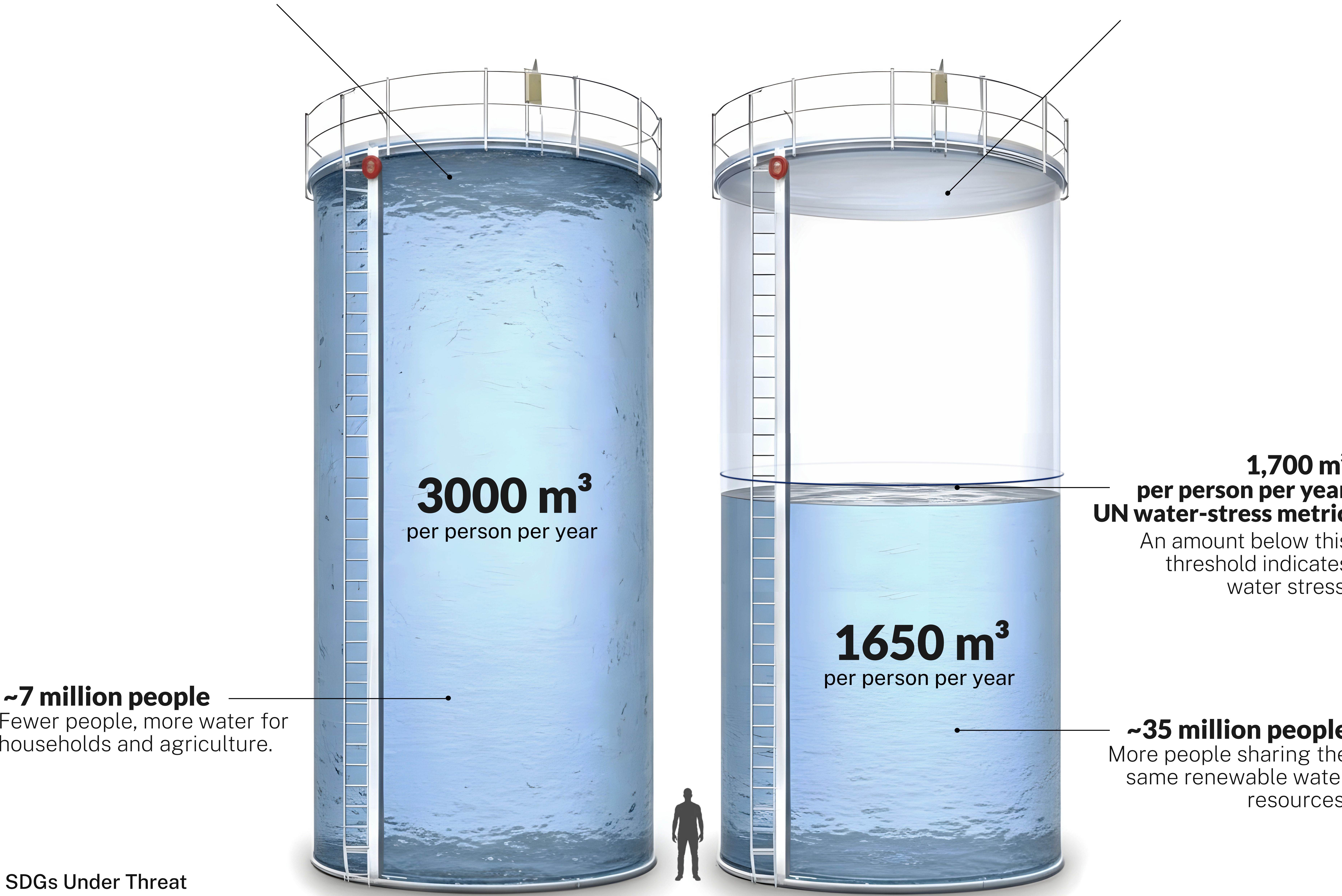
As Ghana’s population has grown, the amount of renewable freshwater available per person has steadily declined. Pollution from illegal mining, urban expansion, and climate stress are accelerating the loss, pushing the country toward water stress.

1960

Ghana had high renewable freshwater availability relative to population size, well above global water-stress thresholds.

2025

Population growth and pressure from illegal mining have reduced per-capita water availability, placing Ghana slightly below the UN water-stress threshold. Forecasts point to continued decline if current trends persist.



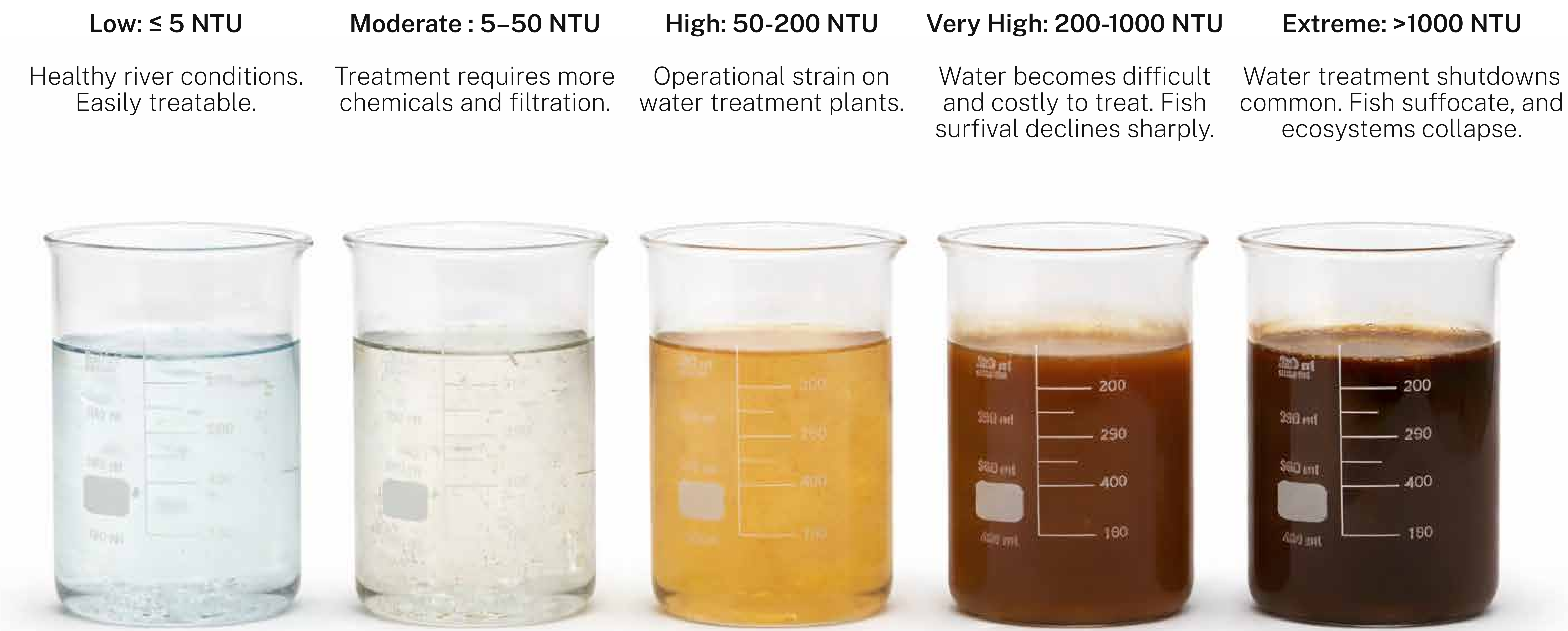
SDGs Under Threat



Muddy Waters

Turbidity levels show how illegal mining is choking Ghana's rivers

Turbidity reflects suspended sediments in water. Galamsey-driven river disturbance has pushed turbidity in several Ghanaian rivers far above treatment thresholds, increasing water losses, ecosystem stress, and supply shutdowns.

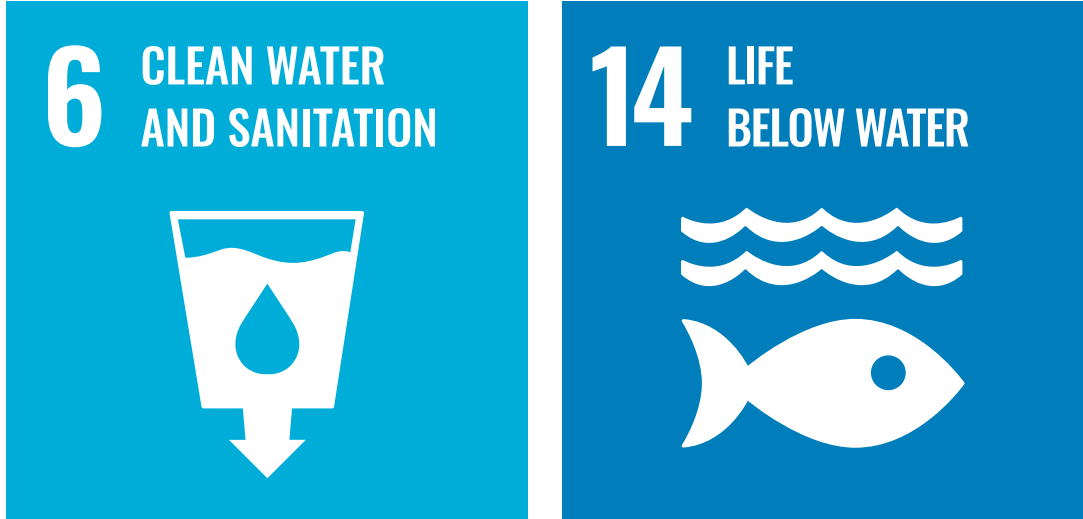


Galamsey-affected rivers (NTU)

• Black Volta (Bamboi): 4.8	• Offin (Barekese): 18.2	• Tano(Elubo): 834	• Ayensu: 94,800
	• Densu (Nsawam): 19.1	• Bia (Dadieso): 372	• Birim: 64,000
	• Densu (Mangoase): 44		• Ankobra: 2,920

Although enforcement efforts reduced turbidity by 70–90% at monitored sites such as Densu at Mangoase (from 256 NTU to 44 NTU year-on-year), levels in most galamsey-affected rivers remain above safe treatment thresholds. As a result, at least 5 Ghana Water Company Limited plants drawing from rivers like Pra, Bonsa, and Birim have faced shutdowns, leaving gains fragile without sustained control.

SDGs Under Threat



Turbidity levels fluctuate seasonally due to rainfall and mining activity, with these figures representing the latest available as of January 2026 from GWCL/WRC reports

Ghana's Cocoa Legacy In Peril

Beyond water pollution, galamsey is degrading Ghana's cocoa lands. Farm destruction, ageing trees, and disease have cut output by over 50% since 2020, threatening Ghana's position as the world's 2nd largest cocoa producer to Ecuador. The visuals show how galamsey is reshaping three major cocoa regions.

Ashanti Region

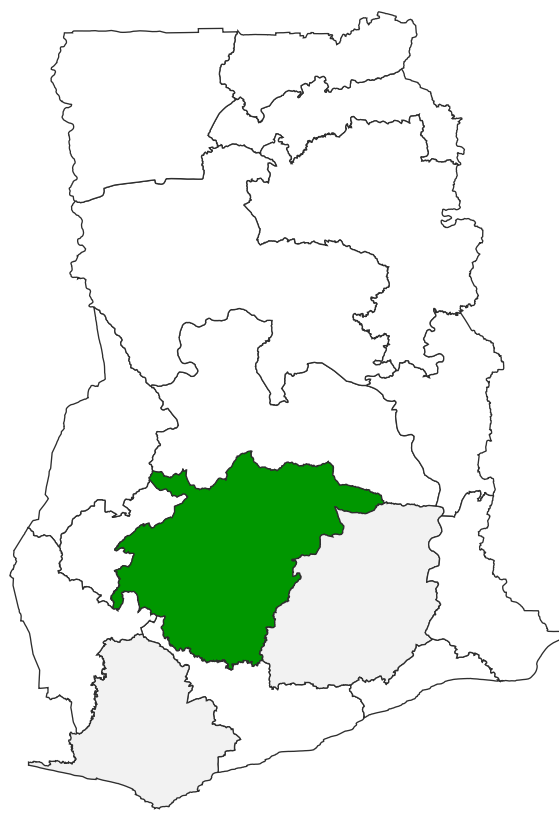
Pollution and soil degradation are reducing yields even where farms remain active.

68%

Of Cocoa farms affected

35.4% ▼

Drop in cocoa production



Western Region

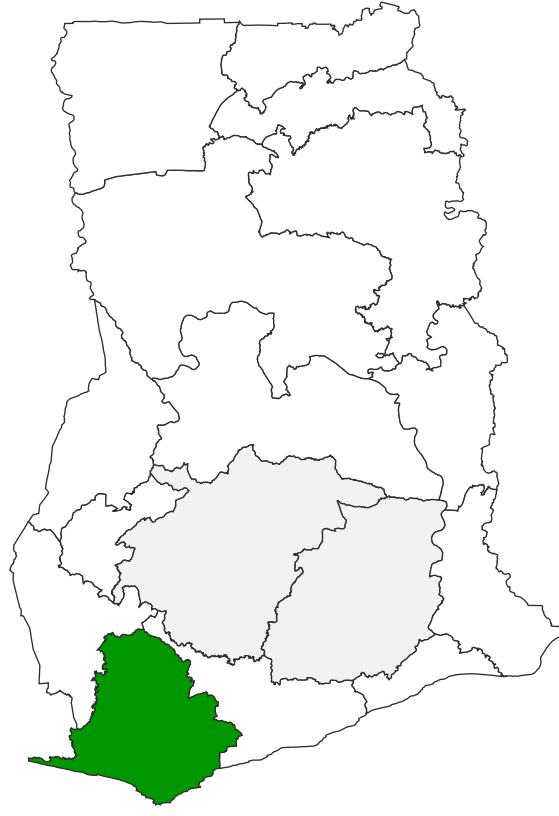
Industrial-scale galamsey is causing the largest contiguous losses of cocoa farmland and livelihoods.

74%

Of Cocoa farms affected

36.5% ▼

Drop in cocoa production



Eastern Region

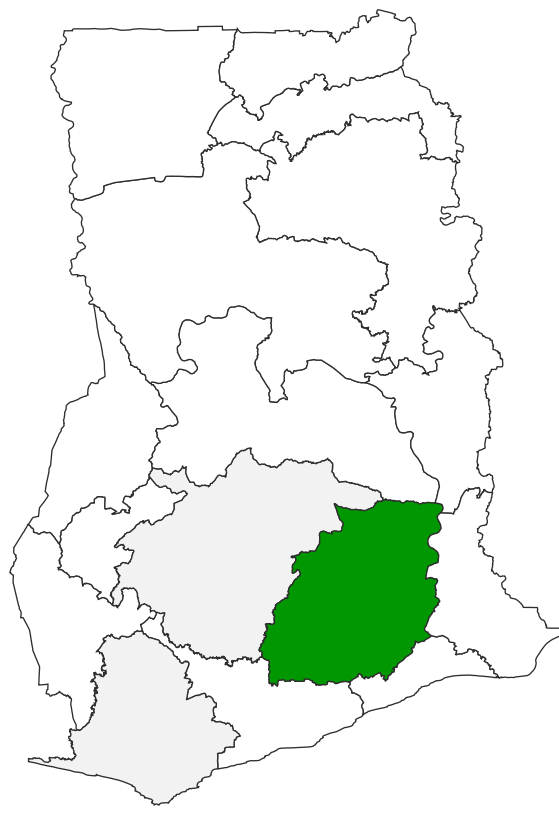
Pollution and soil degradation are reducing yields even where farms remain active.

81%

Of Cocoa farms affected

39.6% ▼

Drop in cocoa production



SDGs Under Threat



Data for 23/24 covers up to June 2024. Forecasts show deeper 24/25 declines from galamsey, old trees, and disease.

Mercury On The Menu

How Toxic Heavy Metals Are Seeping into Ghana's Staple Foodstuffs

The toxins released by illegal mining operations do not dissipate; they are absorbed by staple crops growing in contaminated soil. These toxins also bioaccumulate in river fish, potentially contaminating the entire food chain and turning daily Ghanaian staples into a hidden public health crisis.

This depiction of a popular Ghanaian meal *Ampesi*, illustrates how heavy metals are seeping into the core diet, with risks extending to other meals prepared using the same ingredients, contaminating the entire food basket.

Cassava

Lead levels in cassava from mining areas have been measured at up to **6x** the safety limit, with **mercury** also detected in tubers from active galamsey sites due to uptake from heavily contaminated soils.

River Fish River Fish

High levels of **mercury** have been detected in river fish samples. The WHO warns that consuming contaminated fish is the leading source of mercury ingestion. Lead contamination is also found at **2.8 mg/kg**, a level **7.6x** above safety limits.

Eggs

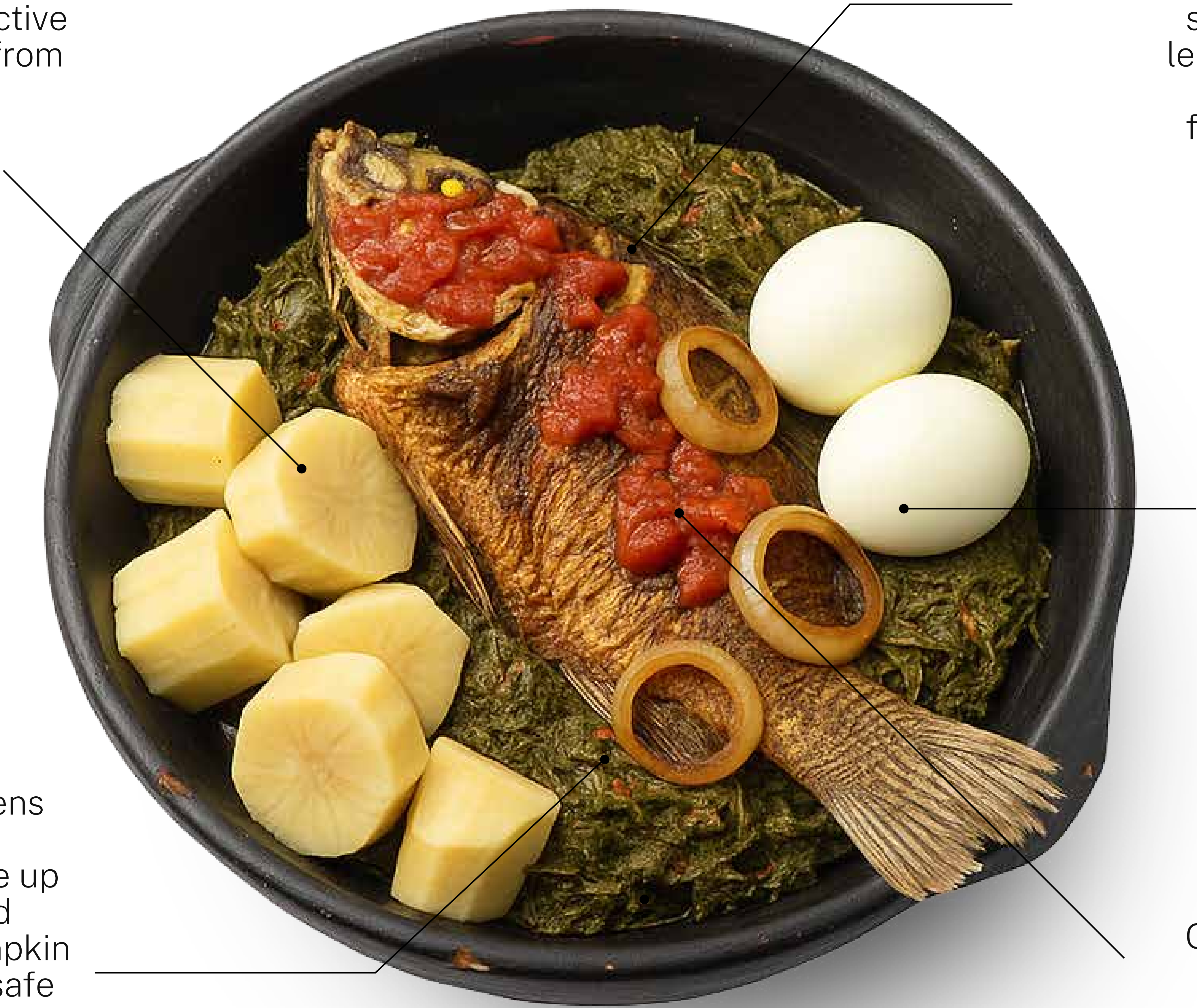
No data shows egg contamination, but feed and water near mining zones remain high-risk.

Leafy Greens

Lead levels in staple leafy greens such as Cocoyam Leaves/Taro (locally called Kontomire), were up to **6.33 mg/kg**, exceeding food safety limits by over **20x**. Pumpkin leave samples also exceeded safe lead limits by **10x**.

Tomatoes

Contained **mercury** levels up to **0.44 mg/kg**, over **10x** food safety limits, and lead levels at **14.4mg/kg**, **48x** the safety limit.



SDGs Under Threat



The Industrial Age Of Galamsey

Once a small-scale, manual activity, galamsey has expanded into a highly mechanized operation. Heavy excavators now clear large areas of farmland and forest, while fleets of Changfang dredges excavate riverbeds and release mercury-laden silt into waterways, contributing to significant levels of water contamination across multiple regions.

\$1+ Billion

Ghana has spent over \$1 billion on excavators between 2013-2024.

1st in West Africa

Ghana is the largest importer of excavators in West Africa and 2nd largest in Africa.

10th Most imported item

Excavators were Ghana’s 10th most imported item in 2024.

89% Increase in imports

Trade data shows excavator imports surged 89% between Oct 2022 and Sep 2024.

6850+ Excavators registered

Authorities have registered over 6,850 excavators in 2025 in a bid to regulate them.

3000 Seized

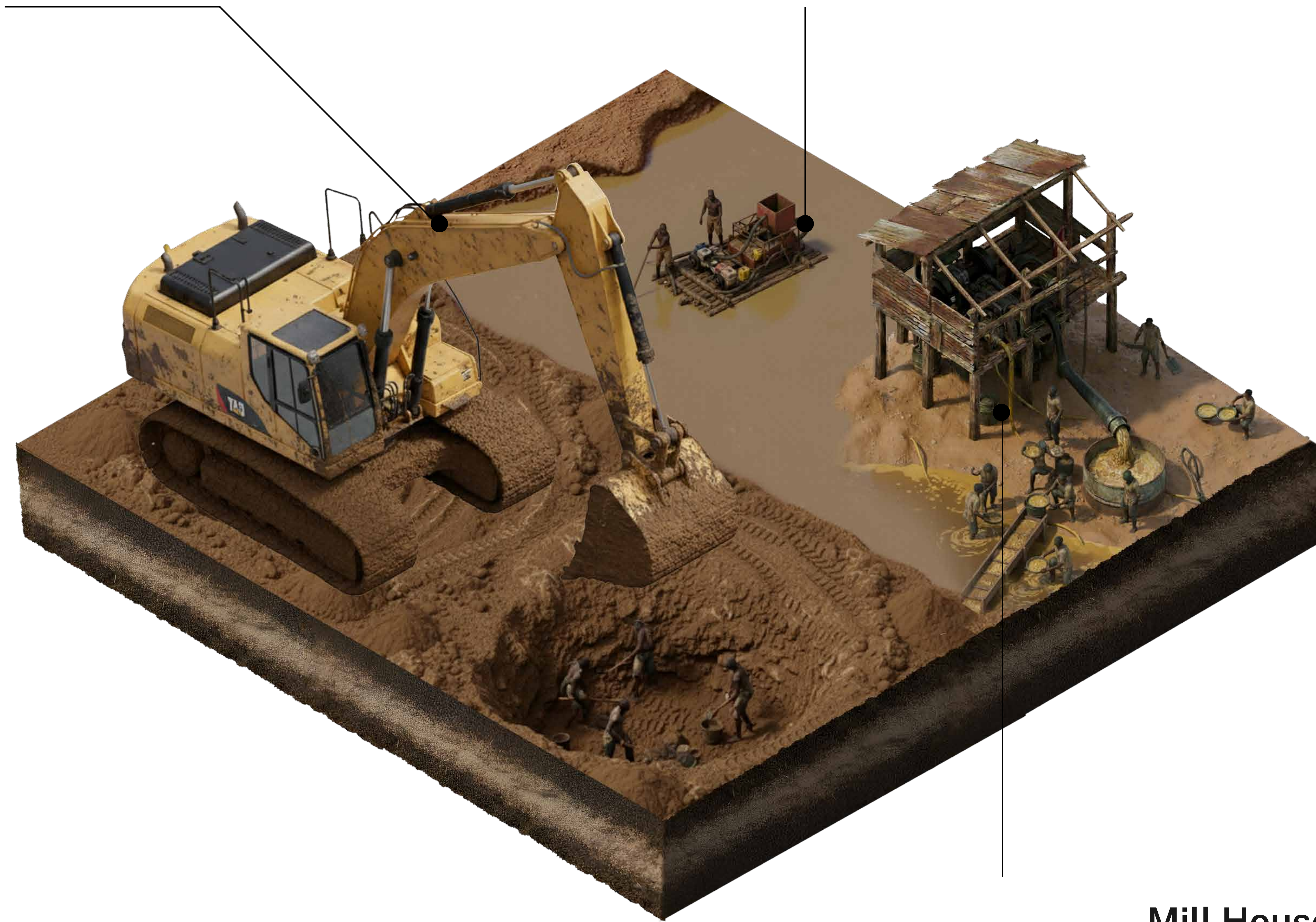
3,000 more unregistered excavators have been seized in anti-galamsey raids.

30,000 - 80,000 m² daily

A single excavator can destroy an area equivalent to 4 to 11 football fields in a day.

Changfang River Dredge

These locally built, diesel-powered machines vacuum riverbeds for gold, mixing mercury onboard and dump toxic waste back into the rivers. Surface water drainage samples from active sites had mercury levels of **82mg/L**, which is **16,400×** over safe limits. A nationwide ban on them took effect in October 2025.



Mill House

These riverside mills process ore from multiple sites and burn mercury-gold mixtures in open air. Surface water drainage samples from active sites had mercury levels of **209 mg/L**, which is **41,800×** over the safe limit for water. This makes them the most toxic hotspots in the entire galamsey chain.

Values represent peak mercury concentrations recorded at active galamsey processing hotspots where contamination is highest-not regional averages

Ministry of Lands and Natural Resources, JoyNews Research, Volza Trade Data, Driver and Vehicle Licensing Authority (DVLA), Ghana Statistical Services Trade Report 2024, Mantey et al. (2020), ScienceDirect

Permission To Proceed

A controversial Legislative Instrument L.I. 2462 - Opened up Forest Reserves to Miners.

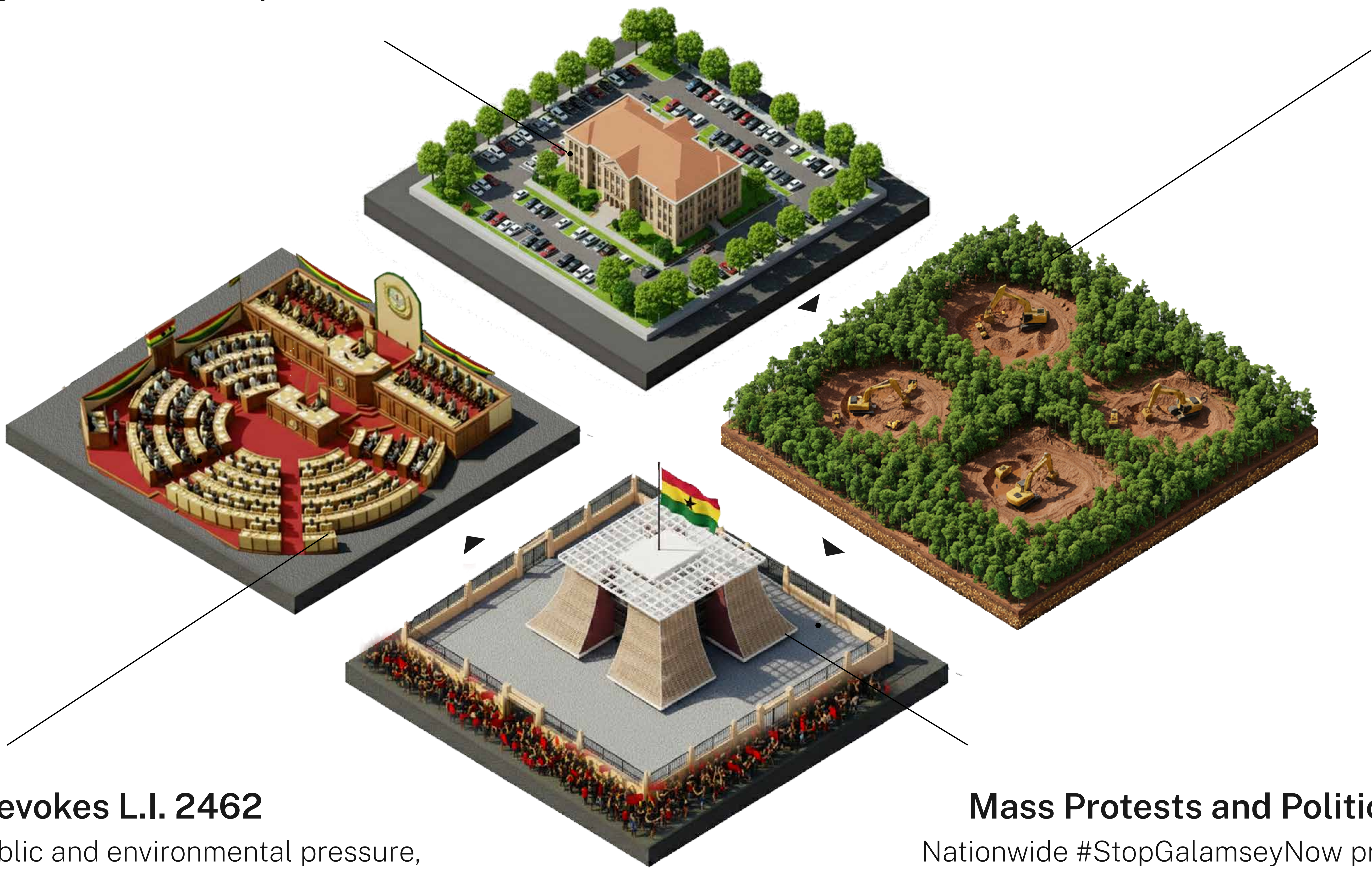
The galamsey crisis is not only the result of illegal activity, but also reflects systemic regulatory failure shaped in part by legal instruments such as the Environmental Protection (Mining in Forest Reserves) Regulations, 2022 (L.I. 2462). Enacted in June 2022, L.I. 2462 permitted mining in forest reserves under specified conditions. Although the regulation has since been repealed, its implementation intensified pressure on protected ecosystems, leaving environmental damage that will persist for decades.

L.I. 2462 Is Passed into Law

L.I. 2462, passed in 2022, permitted mining in forest reserves under specific conditions. This policy shift was compounded by a massive spike in mining licensing: where only **90 licenses** were issued between **1988-2017**, **2,000** more were granted between **Sept. 2017– Jan. 2025**.

Forests Under Siege

Ghana has lost over 44% of its reserves, with more than 120,000 hectares destroyed each year. Poor enforcement allowed galamsey operations to invade reserves, poisoning soil and water with mercury and arsenic.



Government Revokes L.I. 2462

Under growing public and environmental pressure, the government formally revoked L.I. 2462 in December 2025. While the law is now repealed, hundreds of mining licenses issued under its provisions remain active. The government is reviewing these licenses and has already revoked over 300 mining licenses due to regulatory breaches

Mass Protests and Political Change

Nationwide #StopGalamseyNow protests, led by advocacy group Democracy Hub, included demonstrations at Ghana's presidential seat, the Jubilee House. Public outrage over Galamsey and economic turmoil were key factors in the 2024 election, resulting in the ruling party's defeat.

Power, Corruption, And Poverty: The Layers Of Galamsey

Why Ghana Struggles to Break the Galamsey Cycle

Galamsey operates within a network of actors whose differing incentives make enforcement difficult. Financial interests at higher levels and economic pressures at the community level both contribute to the persistence of unregulated mining. Prosecution rates remain low, and security analysts note that the sector's growth raises broader governance and stability concerns.

Alleged Kingpins

Investigations have identified cases where politically connected individuals, traditional authorities, and business figures have been linked to the ownership or financing of illegal mining operations. In one recent example, a regional chairman of an opposition party was charged in connection with an alleged galamsey operation.



Law Enforcement and Authorities

Reports have documented instances of corruption, inconsistent enforcement, and political interference affecting anti-galamsey operations. Investigations and internal disciplinary actions have included cases where security officers were implicated in illicit arrangements with miners, such as the recent dismissal of national security operatives over alleged involvement. These enforcement gaps have coincided with the emergence of armed galamsey groups, creating additional security challenges and complicating law-enforcement efforts.



Vulnerable Communities and the Poor

More than **1 million** Ghanaians work in artisanal and small-scale mining, whose practices increasingly overlap with galamsey on the ground. Another **4 million** depend on the income it provides. In many communities, mining functions as a primary livelihood option amid limited economic opportunities. Daily earnings of **\$70–\$100** far exceed most legal alternatives, pulling many into mining despite the risks.





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About the Author

Fatah Abass - *Data Analyst & Visualization Specialist*

Fatah Abass is a data analyst and visual storyteller based in Ghana, specializing in visual investigations, environmental data stories, and economic reporting. His work has been recognized by Visual Capitalist and he is also a winner of a Maven Analytics international data challenge. He is passionate about data design in the African context, using clear, accessible visuals to democratize complex information, strengthen public understanding, and support evidence-based decision-making across the continent. He has produced data visualizations and reports for clients for clients across business, development, economics, sports, and technology.

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Methodology and Scope

Purpose and Scope

This report presents a data-driven, visual analysis of the environmental and economic impacts of illegal and unregulated gold mining (“galamsey”) in Ghana. It synthesizes publicly available data into a coherent visual narrative focused on measurable system-level impacts rather than individual attribution or allegations.

Research Approach

This investigation synthesizes publicly available data, peer-reviewed research, government reports, and media documentation to examine the scale and impact of illegal mining (galamsey) in Ghana. The work combines quantitative

Data Collection & Verification

All statistics and claims are drawn from verified sources including:

- Peer-reviewed academic journals (e.g., Mantey et al., 2020 in Heliyon; Bempah et al., 2024)
- Ghana government agencies (Water Resources Commission, COCOBOD, Ministry of Lands and Natural Resources)
- International organizations (WHO, UN, Pure Earth)
- Investigative journalism from established Ghanaian media outlets (JoyNews, Graphic Online, Citinewsroom)

Where multiple sources provided data on the same topic, preference was given to official government reports and peer-reviewed studies. All sources are cited throughout the report with direct links provided in the appendices.

Data Visualization

All visuals were produced by the author using manual design and AI-assisted illustration for representational purposes only.

Limitations & Context

Contamination levels cited (e.g., mercury in water, heavy metals in food) represent peak concentrations recorded at active galamsey sites, not regional averages

Turbidity measurements reflect seasonal variations and the latest available data as of January 2026

Intended Use

This report is designed for educational, advocacy, and journalistic purposes. It aims to inform public discourse on environmental governance in Ghana and support evidence-based policy discussions. This report does not attribute legal responsibility, identify specific individuals, conduct original field or laboratory testing, or make clinical health claims. All associations discussed are drawn directly from cited sources.

Sources

Fact	Source	Direct Link
P. 3, Wealth over Water		
Ghana faces potential water import dependency by 2030	GhanaWeb	https://www.ghanaweb.com/GhanaHomePage/NewsArchive/Ghana-may-import-water-from-2030-Lands-Minister-619288
Ghana Water Company wastes up to 50%	Ghana Water Company, Graphic Online	https://www.graphic.com.gh/news/general-news/gwcl-bemoans-cost-of-water-treatment.html
Arsenic (As) found at up to 300x	Ghana A Rocha / Pure Earth	1. https://ghana.arochoa.org/en/2025/10/03/act-now-coalition-tells-president-to-stop-galamsey-ecocide/ 2. https://www.pureearth.org/ghanas-hidden-toxic-legacy-pure-earths-fight-against-heavy-metal-pollution/
Arsenic, Mercury, Lead levels	Baba Imoro Musah et al,	Water concentrations of arsenic, cadmium, lead, and mercury exceeding WHO limits with values of 0.28, 0.13, 0.30, and 0.03 mg/L respectively .. https://pubmed.ncbi.nlm.nih.gov/40562999/
60% of Ghana's water polluted	Water Resources Commission	https://www.business-humanrights.org/en/latest-news/ghana-60-of-water-bodies-polluted-due-to-illegal-mining-and-other-activities-say-authorities/
Producing 1 ounce of gold consumes around 21,000L	Gold.org	https://www.gold.org/about-gold/gold-supply/gold-production/gold-water-benchmarking-report-2024
P.4 More people, Less water		
Per-capita water: 1650m³	Ministry of Works, Housing, and Water Resources	https://www.graphic.com.gh/news/general-news/ghana-news-from-3-000-to-1-650-cubic-meters-per-person-minister-outlines-moves-to-restore-rivers-avert-water-stress.html#:~:text=Ghana's%20per%20capita%20renewable%20water,climate%20commitments%20and%20development%20priorities.
P.5 Muddy Waters		
Turbidity levels: Densu, Offin, Black Volta, Tano, Ankobra, Bia	WRC/GWCL via JoyOnline (Jul 2025 latest)	https://www.myjoyonline.com/improved-turbidity-in-key-rivers-shows-progress-in-galamsey-fight-sammy-gyamfi/
Turbidity in Ayensu river	JoyNews TV report Sep 2025, Ghana Water Ltd.	https://www.youtube.com/watch?v=xhGIC01xcLw&t=18s
Turbidity in Birim river	Abuakwa South Member of Parliament, JoyNews	https://www.myjoyonline.com/abuakwa-south-mp-raises-alarm-over-birim-river-pollution-calls-for-immediate-end-to-galamsey/
P.6, Ghana’s Cocoa Legacy in Peril		
Cocoa exports have fallen by over 50%	Ghana Auditor-General Report / GhanaWeb	https://www.myjoyonline.com/ghanas-cocoa-export-volumes-fall-sharply-by-50-despite-rise-in-earnings-in-2024-auditor-general-report/
Regional damage percentages.	African Agribusiness, COCOBOD	https://africanagribusiness.com/illegal-gold-mining-threatens-cocoa-production-in-ghana/
Regional Cocoa Production Drops	Reuters, COCOBOD	https://www.reuters.com/markets/commodities/ghana-cocoa-output-is-half-average-with-harvest-almost-complete-2024-07-16/
P.7 Industrial Age of Galamsey		
Ghana excavator statistics	JoyNews Research	https://www.myjoyonline.com/ghana-is-the-highest-importer-of-excavators-in-west-africa-2nd-in-africa-joynews-research/
Volza Import Trade data shows excavator imports surged 89%	Volza	https://www.volza.com/p/excavators/import/import-in-ghana/
Authorities register 6,850 excavators	DVLA Ghana	https://www.ghanaweb.com/GhanaHomePage/business/DVLA-registers-6-850-excavators-as-of-July-8-CEO-announces-1991399
3,000 unregistered excavators seized	NewsGhana	https://www.newsghana.com.gh/ghana-seizes-3000-excavators-in-mining-registration-crackdown/
Mining licenses revoked	Citinewsroom	https://www.myjoyonline.com/lands-minister-sets-sept-1-deadline-for-902-small-scale-miners-to-validate-licenses-or-face-revocation/
Ghana loses over \$2 billion annually	CitiNewsRoom	https://citinewsroom.com/2024/09/ghana-losing-about-2bn-annually-to-galamsey-gold-smuggling-uk-govt-raises-alarm/
Alarming Mercury levels in Changfang and Mill site water samples.	Mantey, J., et al. (2020).	[Mill House water contamination (209 mg/L): Table 8; Chamfi/Changfang water contamination (82 mg/L): Table 8] https://www.sciencedirect.com/science/article/pii/S2405844020311567?ref=pdf_download&fr=RR-2&rr=999ea29f1f469339
P.8 Mercury on the Menu		
Mercury levels in tubers	TV3 Ghana / IESS	https://www.facebook.com/TV3GH/posts/iess-research-confirms-dangerous-levels-of-mercury-in-cassava-from-galamsey-site/1173353441493135/
Lead found in river fish and vegetables	Pure Earth / CitiNewsRoom	https://citinewsroom.com/2025/09/toxic-levels-of-lead-found-in-fish-vegetables-across-ghanas-mining-areas-study/
Fish	DailyAccra	https://dailyaccra.com/2025/09/26/heavy-metal-contamination-from-galamsey-puts-ghana-at-risk/
Mercury in Tomatoes	Science Direct	Rebecca Zida Afriyie, et.al Potential health risk of heavy metals in some selected vegetable crops at an artisanal gold mining site: A case study at Moseaso in the Wassa Amenfi West District of Ghana, 4,2023, https://www.sciencedirect.com/science/article/pii/S2773050623000289
P.9 Permission to Proceed		
L.I 2462 analysis	IMANI Africa	https://imaniafrica.org/2025/09/repealing-ghanas-forest-reserve-mining-regulations-navigating-the-path-from-l-i-2462-to-l-i-2501-and-beyond/
L. I2462 Revoked	ModernGhana	https://www.modernghana.com/news/1445382/abraham-koomson-commends-govt-for-revoking-li.html

Appendices

Appendix A: Water Contamination Levels

Compound	Measured Value	Source	Safety Benchmark		Multiplier above safety limit
Arsenic (AS)	3.30 mg/L	Pure Earth	0.01 mg/L (10 µg/L)	WHO, Arsenic Factsheet	330x
Lead (Pb)	Up to 0.3 mg/L	Musah BI, "The impact of artisanal small-scale mining on environmental sustainability in Ghana," (2025), PubMed	0.01 mg/L *No level of lead is safe, and levels should be as low as possible but levels over 0.01 mg/L are cause for concern.	U.S. Environmental Protection Agency (EPA)	30x
Mercury (Hg)	Up to 0.03 mg/L	Musah BI, "The impact of artisanal small-scale mining on environmental sustainability in Ghana," (2025), PubMed	0.001 mg/L	Health Canada. Guidelines for Canadian Drinking Water Quality: Mercury. Government of Canada Publication, 2025.	30x

Appendix B: Vegetable Contamination Levels

Food	Compound	Measured Value	Source	Safety Benchmark		Multiplier above safety limit
Pumpkin Leaves	Lead (Pb)	3.1 mg/kg	Toxic levels of lead found in Fish, Vegetables across Ghana's mining areas – Study By Citi Newsroom	0.3 mg/kg	USDA Foreign Agricultural Service, Global Agricultural information Network	10x
Cocoyam /Taro Leaves	Lead (Pb)	Up to 6.33 mg/kg	Cobbinah, Felix Nana Yaw and Kyei, Sampson Kofi and Azanu, David and S Kabange, Roland, Bioaccumulation Factor and Health Hazards of Heavy Metals in Taro in the Atiwa East Municipality, Ghana. Available at SSRN: https://ssrn.com/abstract=5117863 or http://dx.doi.org/10.2139/ssrn.5117863	0.3 mg/kg	USDA Foreign Agricultural Service, Global Agricultural information Network	21x
Cassava	Lead (Pb)	Up to 1.12 mg/kg	Solomon Odoi Anim, Yaw Gyau Akyereko, Maxwell Adu, William Odoom, Georgina Benewaa Yeboah, Faustina Dufie Wireko-Manu, Quality and safety of cassava roots produced from reclaimed and non-mine sites, Next Research,Volume 2, Issue 2, 2025,100364,ISSN 3050-4759, https://doi.org/10.1016/j.nexres.2025.100364.	0.2 mg/kg	USDA Foreign Agricultural Service, Global Agricultural information Network	6x
Tomato	Lead (Pb)	14.4 mg/kg	Rebecca Zida Afriyie et.al Potential health risk of heavy metals in some selected vegetable crops at an artisanal gold mining site: A case study at Moseaso in the Wassa Amenfi West District of Ghana, Journal of Trace Elements and Minerals, Volume 4, 2023, 100075, ISSN 2773-050	0.3 mg/kg *No level of lead is safe, and levels should be as low as possible but levels over 0.3 mg/kg in vegetables are cause for concern.	https://www.researchgate.net/figure/FAO-WHO-maximum-permissible-values-of-heavy-metals-in-vegetables_tbl1_226944961	48x
	Mercury (Hg)	0.44 mg/kg				10x
River fish	Lead (Pb)	2.28mg/kg	Toxic levels of lead found in Fish, Vegetables across Ghana's mining areas – Study By Citi Newsroom	0.30 mg/kg	Seafish, Maximum permitted levels of lead in fish and fishery products	7.6x

Appendix C: Mining Processing Site Surface Water Contamination Levels

Site/Process	Compound	Measured Value	Source	Safety Benchmark		Multiplier Above Safe Limit
Riverside Mill House	Mercury (Hg)	209 mg/L	Mantey et al.,Mercury contamination of soil and water media from different gold mining operations in Ghana, ScienceDirect (2020). 3. Results 3.1. Concentrations of mercury recorded for each galamsey type across the four environmental media Table 5	0.005 mg/L	Health Canada. Guidelines for Canadian Drinking Water Quality: Mercury. Government of Canada Publication, 2025.	41,800x